

Name (Last, First):

NEU ID:

Assignment 5

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 120 = 3 + 117 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

I. Collecting Data: Experimental Design (11 x 3 = 33 pts)

1) A local hospital compiles data on the length of time a patient is in surgery and the length of their stay in the hospital after surgery. A newbie hospital worker notices that the length of time a patient is in surgery is highly correlated with their hospital stay after. Explain to the newbie why this correlation does not imply a causation.

Answer:

2) The good folks at Apples Inc want to create bags of pre-sliced apples for kids to easily put in their lunch boxes. They want to test different mixtures of preservatives (A, B, and C) on their Honeycrisp apples. They will treat all the apples in a bushel of Honeycrisps by randomly assigning them to each preservation treatment, and then comparing how long they are able to remain in the bag before they begin to brown.

(a) Identify the experimental units, the explanatory and response variables, and the treatments.

Answer:



(b) The researchers plan to use a completely randomized design. Describe how they should assign treatments to the 126 experimental units.

Answer:



(c) The researchers suspect that the type of apple will influence how well the preservation method works. They want to repeat this experiment with 4 different types of apples. Describe how they should change the design of the experiment to account for this addition.

Answer:



3) A study was conducted to determine if taking a daily dose of aspirin reduces the chance of catching the common cold. This study was conducted using 550 volunteers who were not already on an aspirin regimen. The subjects were randomly assigned to one of two groups: a treatment group who received a low dose of aspirin daily, or a control group who received a placebo. The subjects were unaware of what group they were in. At the end of the study, the subjects were asked to meet with a doctor to discuss if they had any symptoms pertaining to the common cold.

(a) Is this study an experiment or an observational study? Explain your answer.

Answer:



(b) What would be the advantage of having this study be double blind?

Answer:



(c) Would blocking according to gender be worthwhile in this study? Explain your answer.

Answer:



(d) Describe what the “placebo effect” would look like in this experiment.

Answer:



4) Type 1 diabetes is thought to be caused by an autoimmune reaction (the body attacks itself by mistake) that destroys the cells in the pancreas that make insulin. It is a disease that currently has no cure. Researchers were to conduct a clinical trial on a promising new drug that helps the body produce its own insulin.

(a) You have 400 volunteers with Type 1 diabetes. Describe a completely randomized design for this experiment.

Answer:

(b) You have 250 children (17 and younger) with Type 1 diabetes (120 females and 130 males), as well as 300 adults (18 and older) with Type 1 diabetes (190 females and 110 males). Describe a completely randomized block design.

Answer:

(c) Explain why a matched pairs design would not work for this experiment.

Answer:



II. Collecting Data: Scope of Inference (5 x 3 = 15 pts)

Directions: For the study below, answer the questions that follow. Then, in the graphic organizer, make a check on the correct scope of the study.

Lack of sleep is associated with increased risk of cardiovascular disease, depression, and other health concerns. A new study now shows that lack of sleep also affects social interactions, making people less willing to help others (empathy). Scientists placed 24 healthy volunteers in a functional magnetic resonance imager (fMRI) to scan their brains after eight hours of sleep and then again after a night of no sleep. The order that they received the treatment of sleep vs no sleep was randomly selected with the flip of a coin. They found that areas of the brain that form the theory of mind network, which is engaged when people empathize with others or try to understand other people's wants and needs, were less active after a sleepless night. "Here, we found that a decrease in the quality of someone's sleep from one night to the next predicted a significant decrease in the desire to help other people from one subsequent day to the next."

Source: <https://www.sciencedaily.com/releases/2022/08/220823143827.htm>

Is this an observational study or an experiment? Explain.										
Does this study involve random sampling or random assignment? Explain.	Which box of inference does this study fall under?									
Can this study make inferences about the population or only the study participants? Explain.	<table border="1"> <tr> <td></td> <td data-bbox="1166 667 1328 867">Inference about cause and effect</td> <td data-bbox="1333 667 1490 867">No Inference about cause and effect</td> </tr> <tr> <td data-bbox="987 873 1161 993">Inference about the Population</td> <td data-bbox="1166 873 1328 993"></td> <td data-bbox="1333 873 1490 993"></td> </tr> <tr> <td data-bbox="987 999 1161 1163">No Inference about the Population</td> <td data-bbox="1166 999 1328 1163"></td> <td data-bbox="1333 999 1490 1163"></td> </tr> </table>		Inference about cause and effect	No Inference about cause and effect	Inference about the Population			No Inference about the Population		
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No Inference about the Population										
Can this study make inferences about cause and effect or just an association between variables? Explain.										

III. Probability Basics (12 x 3 = 36 pts)

Write all probabilities as decimals and round to the nearest three decimal places when needed.

1) You pick 1 marble from a bag that contains 8 marbles (2 blue, 3 red, and 3 yellow). Find the following probabilities.

a) P(Blue)

b) P(Yellow)

c) P(Green)

d) P(Blue or Yellow)

e) P(not Blue)

f) P(Red, Blue, or Yellow)

2) Below are the probabilities of pulling out a particular color from an M&M bag.

- What is the probability of pulling a blue M&M?
- Describe the likelihood of this event.

Brown	Red	Yellow	Green	Orange	Blue
0.30	0.20	0.20	0.10	0.10	?

Answer: a)

b)

3) A survey of 324 people asked what their favorite food was. The results are shown below.

	Pizza	Burgers	Fried Chicken	Other	Total
Less than 18	60	23	5	34	
18 and older	45	33	20	104	
Total					

If we randomly select a person from this sample,

a) What is the probability that person liked fried chicken?

b) What is the probability that person is less than 18 years old and likes burgers?

c) What is the probability that person likes pizza or burgers?

d) What is the probability that person is less than 18 or 18 and older?

IV. The Addition Rule (6 x 3 = 18 pts)

A standard deck of cards has 52 cards. Each deck has 4 different suits: Clubs (black), Spades (black), Diamonds (red), and Hearts (red). Each suit contains the following cards: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and a King.

1) Consider the events listed below and a fair deck of cards.

C = draw a heart

D = draw a face card

E = draw an ace

Which of the following are mutually exclusive? Explain.

a) C and D

b) C and E

c) D and E

2) One six-sided die is rolled. What is the probability that the die will be ...

a) Factor of 12 *or* Factor of 9

b) Less than 3 *or* Greater than 5

3) The probability that a student owns a car is 0.65, and the probability that a student owns a computer is 0.82. If the probability that a student owns both is 0.55, what is the probability that a randomly selected student owns a car or computer?

V. Venn Diagrams, Unions, and Intersections (5 x 3 = 15 pts)

1) At a liberal arts college in the Midwest, 39% of freshmen are enrolled in both a math course and an English course, 11% are enrolled in a math course, but not an English course, and 63% are enrolled in an English course.

a) Construct a Venn diagram to illustrate this situation. Be sure to label all sets.

b) What is the probability that a freshman student selected at random is taking an English course but not a mathematics course?

c) What is the probability a student is not enrolled in either course?

2) Create a table to display all data that is described.

M = Male

F = Female

D = owns a desktop

L = owns a laptop

B = owns both

N = owns neither

$$P(F \cap D) = \frac{21}{235}$$

$$P(M) = \frac{153}{235}$$

$$P(L) = \frac{106}{235}$$

$$P(M \cap L) = \frac{52}{235}$$

$$P(F \cap N) = \frac{6}{235}$$

$$P(N) = \frac{10}{235}$$

$$P(F \cup D) = \frac{86}{235}$$

	Desktop	Laptop	Both	Neither	Total
Female					
Male					
Total					

3) The probability of rain on Saturday or Sunday is given as 0.5. The probability of rain on Saturday is 0.2 and the probability of rain on Sunday is 0.4. What is the probability that it will rain on both Saturday and Sunday?

THE END